

Two Approaches to the Riemann Hypothesis

Beurling–Nyman and Connes

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Independent

Abstract

Two approaches to the Riemann Hypothesis reformulate it as a spectral problem, each starting from a different space. The Beurling–Nyman (BN) approach works in $L^2(0, 1)$: RH is equivalent to the completeness of the BN functions $\{\rho_{1/k}\}$, or equivalently to $d_N \rightarrow 0$. The Connes approach works in $L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*)$: RH is equivalent to a spectral property of the scaling operator on the adèle class space, where the Riemann zeros appear as the absorption spectrum (missing eigenvalues). Both approaches translate RH into a statement about a Hilbert space and an operator. Both reach the same wall: the zeros of ζ on or off the critical line. They differ in what they make visible: BN illuminates the prime structure (explicit formula, rate, pair correlation); Connes illuminates the global arithmetic (the explicit formula as a trace formula, the connection to the Weil conjectures over finite fields). Neither has crossed the wall.

1 The Beurling–Nyman Approach

Definition 1 (BN setup [1, 2]). The BN functions: $\rho_{1/k}(x) = \{1/(kx)\} - (1/k)\{1/x\}$ for $k \geq 2$. The span: $\mathcal{V}_N = \text{span}\{\rho_{1/2}, \dots, \rho_{1/(N+1)}\} \subset L^2(0, 1)$. The BN distance: $d_N^2 = \inf_{f \in \mathcal{V}_N} \|1_{(0,1)} - f\|_{L^2(0,1)}^2$.

Theorem 2 (Nyman–Beurling–Báez-Duarte). $d_N \rightarrow 0$ as $N \rightarrow \infty$ if and only if RH holds.

Remark 3 (What BN makes visible). The BN distance connects to:

- *Primes*: $d_N^2 \asymp \int_N^{2N} |E(e^u)|^2 / e^{2u} du$, where $E(x) = \psi(x) - x$ is the prime counting error.
- *Gram matrix*: $G_N \approx DT_N(\sigma)D$ with $\sigma(t) = |\zeta(1/2 + it)|^2 / (1/4 + t^2)$.
- *Pair correlation*: GUE statistics of zeros control the BN angle sum.
- *Rate*: unconditionally, $d_N^2 = O(\exp(-c(\log N)^{1/3}))$.

The BN approach is explicitly arithmetic: the basis functions $\rho_{1/k}$ are built from fractional parts of $1/(kx)$, encoding the multiplicative structure of the integers.

2 The Connes Approach

Definition 4 (Adèle class space [3]). The ring of adèles: $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p \mathbb{Q}_p$ (restricted product, with $\mathbb{Z}_p$ at almost all primes). The idèle class group: $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^* / \mathbb{Q}^*$. The adèle class space: $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}} / \mathbb{Q}^*$ (additive quotient, not a group). The Hilbert space: $\mathcal{H} = L^2(C_{\mathbb{Q}}, d^*u)$ where d^*u is the multiplicative Haar measure on $C_{\mathbb{Q}}$.

Theorem 5 (Connes' explicit formula as trace [3]). *The scaling operator $U_{\lambda} : f(u) \mapsto f(\lambda u)$ on $\mathcal{H}$ generates a one-parameter group with infinitesimal generator D . The Weil explicit formula:*

$$\sum_{\rho} h(\rho) = h(1) + h(0) - \sum_p \sum_{n \geq 1} \frac{\log p}{p^{n/2}} (\hat{h}(n \log p) + \hat{h}(-n \log p)) - \dots$$

(for a suitable test function h) is the trace formula for U :

$$\text{Tr}(f(D)) = \sum_{\rho} f\left(\frac{\rho - 1/2}{i}\right) + (\text{boundary terms}),$$

where the sum is over nontrivial zeros ρ of ζ .

Remark 6 (The zeros as absorption spectrum). In a standard spectral problem, eigenvalues appear as peaks of the spectral measure. In Connes' construction, the zeros appear as the *absorption spectrum*: frequencies missing from the spectral measure of D relative to what the trace formula predicts. The analogy: the zeros of ζ are where the primes “absorb” the spectrum, exactly as atomic absorption lines correspond to electronic transitions. RH would say: the absorption lines all lie on a single frequency ($\text{Re}(s) = 1/2$).

Definition 7 (The spectral realization problem [3]). The *Hilbert–Pólya program*: find a self-adjoint operator T on a Hilbert space such that the nontrivial zeros of ζ are $\rho_k = 1/2 + i\lambda_k$ where λ_k are eigenvalues of T . RH would follow from the spectral theorem (T self-adjoint $\Rightarrow \lambda_k \in \mathbb{R}$). Connes' operator D is almost this: it produces the zeros as absorption spectrum, but not as eigenvalues in the usual sense. The precise open problem: find a suitable Sobolev completion of $\mathcal{H}$ in which the absorption spectrum becomes a genuine spectrum and D is essentially self-adjoint.

3 Comparison

Theorem 8 (What the approaches share). *Both approaches:*

1. *Translate RH into a Hilbert space statement. BN: completeness in $L^2(0, 1)$. Connes: spectral property of D on $L^2(C_{\mathbb{Q}})$.*
2. *Use the explicit formula as a key tool. BN: $d_N^2 \asymp \int |E|^2 / e^{2u} du$ via the prime counting error. Connes: the Weil explicit formula is the trace formula for D .*
3. *Reach the same wall. BN: whether the off-critical residual $d_{\infty}^2 = 0$. Connes: whether the absorption spectrum lies on $\text{Re}(s) = 1/2$. These are the same statement.*

4. *Cannot cross the wall with their internal tools.* BN cannot close the non-Toeplitz correction without new input. Connes cannot achieve the spectral realization without a new Sobolev theory.

Theorem 9 (What the approaches make distinct). *1. **BN makes prime structure visible.** The Gram matrix, pair correlation, Vinogradov–Korobov rate, and numerical computation of d_N are all accessible via BN. The approach is explicit and computable.*

2. ***Connes makes global arithmetic visible.** The Weil conjectures over finite fields provide the exact template: for a curve over $\mathbb{F}_q$, the analogous trace formula holds and the analog of RH was proved by Weil (1948) and Deligne (1974). The Connes approach frames RH as the $\mathbb{Q}$ case of a general theorem that holds over every other field. The analogy: $\mathbb{Q}$ is the hardest base field.*
3. ***The scaling is different.** BN lives at the scale of individual primes and integers. Connes lives at the global scale of the adèle class space. The two approaches look at the same object (the zeros of ζ) through lenses of very different magnification.*

4 The Weil Conjectures as Template

Theorem 10 (Weil (1948), Deligne (1974) [4, 6]). *Let X be a smooth projective variety over $\mathbb{F}_q$. The zeta function $Z(X/\mathbb{F}_q, T) = \exp(\sum_{n \geq 1} \#X(\mathbb{F}_{q^n})T^n/n)$ satisfies:*

1. *Rationality: Z is a rational function of T .*
2. *Functional equation: $Z(T) = \pm q^{x/2} T^x Z(1/q^d T)^{-1}$.*
3. *Riemann hypothesis: the zeros of Z have absolute value $q^{-k/2}$ for explicit integers k (the zeros of the k -th cohomology factor have absolute value $q^{-k/2}$).*

Deligne proved (3) via the theory of étale cohomology.

Remark 11 (What the Weil conjectures say for RH over $\mathbb{Q}$). Over $\mathbb{F}_q$, the analog of RH is a theorem. The key input was algebraic geometry: the Frobenius acting on étale cohomology is known to have the right eigenvalues by the theory of weights. For $\mathbb{Q}$: the analog of Frobenius is the prime distribution. The analog of étale cohomology does not exist (or has not been constructed). The Connes approach is an attempt to build the $\mathbb{Q}$ analog. The wall is exactly the step where the analogy runs out.

5 Two Walls, One Zero

Theorem 12 (The wall is the same wall). *The BN wall and the Connes wall are the same:*

$$d_\infty^2 = 0 \iff \text{absorption spectrum of } D \subset \{s : \operatorname{Re}(s) = 1/2\} \iff \text{RH.}$$

The BN formulation reaches the wall via prime counting, Gram matrices, and functional analysis. The Connes formulation reaches the wall via adèles, trace formulas, and noncommutative geometry. They approach from opposite directions and stop at the same place.

Remark 13 (What two approaches mean). Two independent approaches reaching the same wall is evidence that the wall is a real obstruction, not a failure of imagination. It is not that we have not been creative enough. It is that we have been creative in two orthogonal directions and reached the same boundary. The wall is not behind us; it is in front of both approaches at once.

References

- [1] Rincón, D. (2026). The Riemann Hypothesis: A Synthesis. Preprint.
- [2] Rincón, D. (2026). Wisdom. Preprint.
- [3] Rincón, D. (2026). Connes: Noncommutative Geometric Approach to RH. Preprint.
- [4] Rincón, D. (2026). The Weil Conjectures. Preprint.
- [5] Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica*, 5(1), 29–106.
- [6] Deligne, P. (1974). La conjecture de Weil I. *Publications Mathématiques de l’IHÉS*, 43, 273–307.